ECMWF
EUROPEAN CENTRE FOR MEDIUM-RANGE WEATHER FORECASTS

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iFAB INTERNATIONAL FOUNDATION BIG DATA AND ARTIFICIAL INTELLIGENCE FOR HUMAN DEVELOPMENT

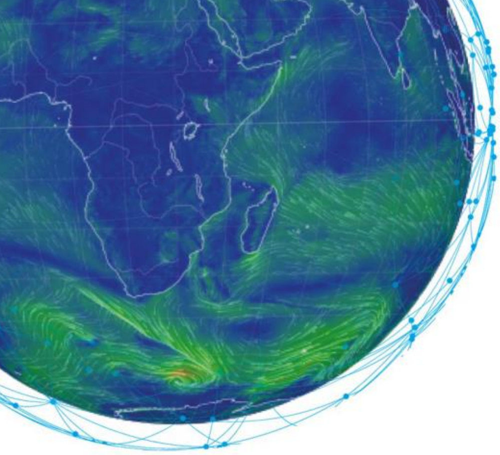
MOOC
Machine Learning in Weather & Climate

TRANSCRIPT

Concept and state-of-the-art

Expert: Peter Dueben

University of Luxembourg
Competence Centre



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Introduction

Hello and welcome to the module Forecast Model. To kick off this module, I would like to introduce you to our next speaker, Peter Dueben. Peter is the Head of the Earth System Modelling Section at the European Centre for Medium-Range Weather Forecasts and will explain to us why it is actually a good idea to consider machine learning for weather and climate modelling.

Peter, the floor is yours.

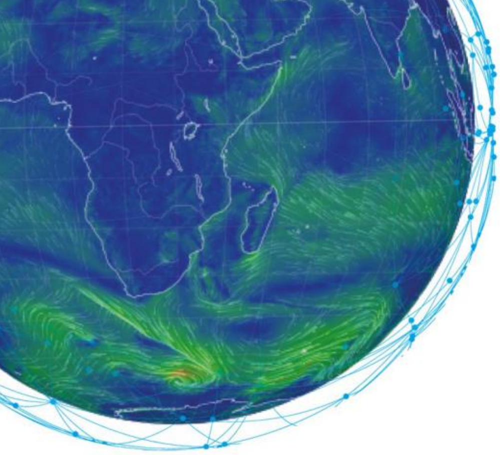
Script

As they are used for weather and climate predictions, Earth system models are fascinating engineering tools. They enable us to predict weather a couple of days ahead, identify large scale weather anomalies – for example, whether the winter in Europe is likely to be warmer or colder than usual – up to months ahead, and even allow to make projections about a change of the climate given a certain amount of CO2 which is emitted into the atmosphere.

But how do these models work and how are they built? – There are five main ingredients.

First, you need equations that describe the motion in the different components of the Earth system. For atmosphere and ocean, the motion is described by equations that are valid for fluids such as air or water. The equations are derived from physical reasoning and assumptions such as the conservation of mass, or the basic laws of thermodynamic. In principle, the equations should be sufficient to describe the dynamics of the Earth system exactly. However, there is a major problem. Unfortunately, we are not able to solve the equations analytically as they have so-called non-linear terms. We can only solve the equations in an approximate way. To do this, we need our second ingredient.

To solve the equations, we discretise the space into grid-points. The main properties of the flow are not described continuously anymore. In contrast, they are only described at the points of a numerical grid. The good news is that we can now solve the equations of motion with a numerical model. The bad news is, that we are not able to resolve features of the Earth system explicitly, which are smaller than the spacing between the grid points.



However, to make the calculation of a future state of the Earth system possible, a lot of efforts need to be made to find the best so-called discretisation of the equations of motion on a given grid. A discretisation is the way how the variables on the different grid-points interact with each other in order to follow the motion that is described by the equations. Many different options are available how to do this and each of them will have different advantages and disadvantages. Applied mathematicians are kept busy by the quest for the most accurate and most efficient method when doing the calculations, and it takes more than 10 years to build a new model for the atmosphere or ocean. There is often a trade-off between accuracy and performance. Cheap and efficient methods allow for the use of finer grids and therefore higher resolution of the models. However, numerical errors can cause instabilities or bad forecasts.

The grids which are used in state-of-the-art Earth system models have hundreds-of-millions of grid-points which are distributed in a horizontal layer over the globe. They also have more than a hundred of these layers in the vertical direction. While we can solve the equations on the grid, there are still billions of operations which need to be performed to update the state of the Earth system by a couple of minutes. This is impossible to do by humans. A supercomputer is needed and forecast centres such as ECMWF host supercomputers which are among the fastest computers in the world.

The final ingredient results from the limit in resolution of the models. Many features of the Earth system, such as individual clouds, are too small to be represented on the grid. However, they can still be very important – just imagine an atmosphere without clouds. To represent those features, so-called sub-grid-scale parametrisation schemes are developed. The schemes are using information on the grid – such as temperature and humidity – to approximate what sub-grid features would be doing if they were resolved. This process is important but can always only be a rough approximation of the actual physical process. Parametrisation schemes are therefore very difficult to design, and many domain scientists are required to attain the best solutions for small-scale dynamics of clouds, convection, processes at the land surface or sea-ice, and small turbulent eddies.

Summary

I find it really fascinating how the different “ingredients” are put together to address such a difficult task. I am sure not everyone who is using weather forecasts in their daily life would be aware of the complexity behind the scenes.

Many thanks Peter for this interesting overview on how conventional weather and climate models are developed. If you have any question or would like to discuss this subject in further detail, please use our forum on Moodle. It will be a pleasure to hear you!